

Department of Computer Science and Engineering
National Institute of Technology Karnataka, Surathkal – 575025

Course Name : Compiler Design Lab

Course Code : CS304

Lab Number : Week 3 (S2)

Experiment Title : **Lexical Analysis using Lex**

Question 1: Write a **Lexical Analyzer** using **Lex** that scans a C++ source program and identifies the following lexical tokens:

- Keywords
- Identifiers
- Numeric Constants (decimal, octal, hexadecimal, and floating-point)
- Operators (Arithmetic, Relational, Logical, Assignment)
- Delimiters

The program should generate and display a **Token Table** containing:

- Lexeme
- Token Category
- Line Number
- Frequency of Occurrence

Question 2: Write a **Lexical Analyzer** using **Lex** that identifies all **variables** and **function prototypes/definitions** present in a C++ source program.

The program should construct and display a **Symbol Table** containing:

- Variable/Function Name
- Category (Variable/Function)
- Data Type / Return Type
- Scope (Global/Local)
- Number of Parameters (for functions)
- Line Number

Code and Results Submission Deadline : Between 11:30 – 11:45 AM on the Lab Day

****